

Chapter Summary

Key Terms

4.1 Hindu-Arabic Positional System

- numeral
- number
- exponential expression
- base
- exponent
- place value
- base 10 system
- Hindu-Arabic numeration system
- expanded form

4.2 Early Numeration Systems

- additive system of numbers
- positional system of numbers
- Babylonian system of numbers
- Mayan system of numbers
- Roman system of numbers

4.3 Converting with Base Systems

- base 10
- remainder
- dividend
- divisor
- quotient

Key Concepts

4.1 Hindu-Arabic Positional System

- Exponents are used to represent repeated multiplication of a base.
- In arithmetic, exponents are computed before multiplication, division, addition, and subtraction. Computing an exponent is done by multiplying the base by itself the number of times equal to the exponent.
- The system of numbers currently used is the Hindu-Arabic system. Digits in this system take on values based on their place in the number. The place values are determined by multiplying the digit by 10 raised to the appropriate power.
- The expanded form of a Hindu-Arabic number is the sum of each digit times 10 raised to the exponent for that place value.

4.2 Early Numeration Systems

- Historically, there have been many systems for numbering. One system is an additive system, in which symbols are repeated to express larger numbers. Another system is a positional system, in which the digits and their positions determine the quantity being represented.
- The Babylonian system was a combination of a positional and additive system. It used 60 as its base. Using that in the positional system makes it possible to convert between Babylonian and Hindu-Arabic numbers.
- The Mayan system was a combination of a positional and additive system. It used 20 as its base. Using that in the positional system makes it possible to convert between Mayan and Hindu-Arabic numbers.
- The Roman system was an additive system. Knowing what each symbol represents makes it possible to convert between Roman and Hindu-Arabic numbers.

4.3 Converting with Base Systems

- The system we use is the base 10 system. Base 10 is not the only base that can be used. To use another base, one could start with a list of numbers in that base.
- To indicate that a number is written in a base other than 10, a subscript is appended to the end of the number. That subscript indicates the base for the number.
- Numbers written in a base smaller than 10 use the same symbols as base 10. However, when using bases larger

than 10, the symbols A, B, C, ... are used to represent digits larger than 9.

- To convert from a number written in a base other than 10 into a base 10 number, the number is written in expanded form and then that expression is computed.
- To convert a number from base 10 into another base, the base 10 number is repeatedly divided by the new base. The remainders when performing these divisions become the digits for the number in the new base.
- Common errors can be detected when performing base conversions.

4.4 Addition and Subtraction in Base Systems

- Addition tables for bases other than 10 can be built using the same processes that are used in base 10, including using a number line.
- Addition in bases other than base 10 use the same processes as addition in base 10, but use the addition table for that base.
- Subtraction in bases other than base 10 use the same processes as subtraction in base 10, but use the addition table for that base.

4.5 Multiplication and Division in Base Systems

- Multiplication tables for bases other than 10 can be built using the same processes that are used in base 10, including using repeated addition and the addition table for the base.
- Multiplication in bases other than base 10 use the same processes as multiplication in base 10, but use the multiplication table for that base.
- Basic division in bases other than base 10 use the same processes as basic division in base 10, where the missing factor process is used.

Videos

4.1 Hindu-Arabic Positional System

- [Exponential Notation \(https://openstax.org/r/Exponential_Notation\)](https://openstax.org/r/Exponential_Notation)

4.2 Early Numeration Systems

- [Converting Between Babylonian and Hindu-Arabic numbers \(https://openstax.org/r/Babylonian_to_Hindu-Arabic_Numbers\)](https://openstax.org/r/Babylonian_to_Hindu-Arabic_Numbers)
- [Converting Mayan Numbers to Hindu-Arabic Numbers \(https://openstax.org/r/Mayan_to_Hindu-Arabic_Numbers\)](https://openstax.org/r/Mayan_to_Hindu-Arabic_Numbers)
- [Converting From Roman Numbers to Hindu-Arabic Numbers \(https://openstax.org/r/Roman_to_Hindu-Arabic_Numbers\)](https://openstax.org/r/Roman_to_Hindu-Arabic_Numbers)
- [Converting From Hindu-Arabic Numbers to Roman Numbers \(https://openstax.org/r/Hindu-Arabic_to_Roman_Numbers\)](https://openstax.org/r/Hindu-Arabic_to_Roman_Numbers)

4.3 Converting with Base Systems

- [Convert Base 7 to Base 10 \(https://openstax.org/r/Convert_Base_7_to_Base_10\)](https://openstax.org/r/Convert_Base_7_to_Base_10)
- [Converting from Base 10 to Another Base \(https://openstax.org/r/Base_10_to_Another_Base\)](https://openstax.org/r/Base_10_to_Another_Base)

Projects

Additive Systems

Go online. Google “additive number systems.” What system comes up?

- Describe the additive system you found.

Using Google, identify three more additive systems of numbers.

- Compare and contrast the systems you found. For instance, how many times can a symbol be used before a new symbol is used.
- Identify three situations where additive systems are still used.

Computers and Bases

Use Google to determine what base computers use.

Were other bases attempted for use in computers?

Determine why the base used in computers is appropriate.

Determine how the base used in computers is related to the circuitry in computers.

Determine how Boolean logic and the base used in computers are related, and might be identical.

There is research into using qubits in computers. Find out what qubits are and how can they improve computing speed.

Cultures Using Base Systems Other Than 10

Using Google, find three cultures, other than Babylonian or Mayan, that use base systems other than 10.

- Tell what base is used for each system.
- If possible, determine why the culture used that base system.
- Choose one of those systems. Explain that base system. Be sure to address whether the system is additive, place-value based, a blend of the two, and if it employs a zero.

History of Zero

Using any resources available to you, determine the history of 0 in at least three different numbering systems. Address at least when and why such a development occurred and why a 0 is vital to the use of a positional system.

Numbering Systems from Other Global Regions

Using any resources available to you, find at least three numbering systems from sub-Saharan Africa, Australia, China, or the Pacific Islands. Explore if they are positional or additive systems (or combinations!), the terminology of the system, if they used a 0, and what base they employed (if positional).

Chapter Review

Hindu-Arabic Positional System

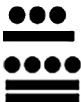
1. What is the base of 5^7 ?
2. What is the exponent of 5^7 ?
3. Compute $4^5 + 3^2$.
4. Convert the Hindu-Arabic number into expanded form: 4,201.
5. Convert the expression to a Hindu-Arabic numeral: $6 \times 10^3 + 3 \times 10^2 + 7 \times 10^1 + 5 \times 10^0$.

Early Numeration Systems

6. Which systems—Hindu-Arabic, Roman, Mayan, or Babylonian—are additive systems?
7. Which systems—Hindu-Arabic, Roman, Mayan, or Babylonian—are positional systems?
8. Which systems—Hindu-Arabic, Roman, Mayan, or Babylonian—use a 0?
9. In the Babylonian system, what are the place values based on?
10. In the Mayan system, what are the place values based on?
11. Convert the Babylonian numeral to a Hindu-Arabic numeral.



12. Convert the Mayan numeral to a Hindu-Arabic numeral.



13. Convert the Roman numeral MMCDXLVII into a Hindu-Arabic numeral.
14. Convert the Hindu-Arabic numeral 394 to a Roman numeral.

Converting with Base Systems

15. List the numbers from 0 to 100 in base 5.
16. In base 8, what is the place value of the 3 in the number 638_8 ?
17. How many symbols are needed for a base 17 system?
18. What does it mean for a number to be in base 6?
19. What symbols are used in a base 12 system?
20. When converting from a base 10 number to a base 2 number, would the number of digits decrease?
21. Convert 311_5 to base 10.
22. Convert 45_{12} to base 10.
23. Convert 1001_2 to base 10.
24. Convert 459 to base 8.
25. Convert 1198 to base 12.
26. Convert 38 to base 2.
27. When converting from base 10 to base 4, the result obtained was 142_4 . How can you tell an error was made?

Addition and Subtraction in Base Systems

28. Create the addition table for base 5.
29. How many addition rules are there for a base 7 system?
30. Calculate $34_5 + 44_5$.
31. Calculate $A7_{12} + 88_{12}$.

32. Calculate $541_6 - 233_6$.
33. Calculate $5B_{12} - 1A_{12}$.
34. When adding in base 8, the result 911_8 is found. How do we know a mistake was made?

Multiplication and Division in Base Systems

35. What is the process for creating the multiplication table for a base other than 10?
36. Create the multiplication table for base 7.
37. How many multiplication rules are there in a base 3 system?
38. Calculate $14_7 \times 25_7$.
39. Calculate $67_{12} \times 3B_{12}$.
40. Calculate $42_7 \div 5_7$.
41. Calculate $38_{12} \div B_{12}$.
42. When multiplying $14_{12} \times 10_{12}$, the result 140 is found. How do we know a mistake was made?

Chapter Test

1. Expand the Hindu-Arabic numeral 5,789.
2. Evaluate the expression $4 \times 7^3 + 5 \times 2^5$.
3. Rewrite $6 \times 10^5 + 0 \times 10^4 + 8 \times 10^3 + 0 \times 10^2 + 1 \times 10^1 + 9 \times 10^0$ in Hindu-Arabic form.
4. Convert the Babylonian numeral to a Hindu-Arabic numeral.



5. Convert the Mayan number to a Hindu Arabic numeral.



6. What base system did the Babylonians use?
7. Which system—Roman, Babylonian, Mayan—used place values?
8. Convert the Roman numeral MDXLVII to a Hindu-Arabic numeral.
9. How many symbols are needed for a base 9 system?
10. For a system in a base larger than 10, what symbols are used as digits representing more than 10?
11. Convert 132_8 to a base 10 number.
12. List the numbers in base 4 up to 100_4 .
13. Convert 74 to a base 12 number.
14. Create the addition table for base 4.
15. Calculate $314_6 + 453_6$.
16. Calculate $4B1_{12} - 2A6_{12}$.
17. When calculating $23_6 + 53_6$, a student obtains 76_6 . How do you know an error was made?
18. Create the multiplication table for base 4.
19. Calculate $323_4 \times 132_4$.
20. Calculate $21_4 \div 3_4$.